

Chapter 7, Problem 33P

Problem

The voltage across a 10-mH inductor is $45\delta(t - 2)$ mV. Find the inductor current, assuming that the inductor is initially uncharged.

Step-by-step solution

Step 1 of 2

The expression for the current flowing through the inductor is,

$$i(t) = \frac{1}{L} \int_0^t v(t) dt + v(0) \dots\dots (1)$$

Here, $v(t)$ is the voltage across the inductor.

Step 2 of 2

As, the inductor is initially uncharged,

$$v(0) = 0 \text{ V}$$

Recall equation (1).

$$i(t) = \frac{1}{L} \int_0^t v(t) dt + v(0)$$

Substitute 10×10^{-3} for L , $45\delta(t - 2) \times 10^{-3}$ for $v(t)$, and 0 for $v(0)$.

$$i(t) = \frac{1}{10 \times 10^{-3}} \int_0^t 45\delta(t - 2) \times 10^{-3} dt + 0$$

$$i(t) = 4.5u(t - 2) \text{ A} \quad \left(\because \int \delta(t) dt = u(t) \right)$$

Therefore, the inductor current is $\boxed{4.5u(t - 2) \text{ A}}$.